

A Simple WIRELESS CHARGER For Smartphones

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Smartphone batteries need to be charged frequently, generally more than once a day. But their charging cables tend to get damaged after some time due to excessive use, which can be annoying—especially if they fail when these are needed the most. A wireless charger can be a convenient alternative; you simply set it up and forget it. But how does it work, exactly?

While wireless charging may seem like a recent invention, its origin dates to more than a hundred

BILL OF MATERIAL

Component	Quantity	Purpose
0.3 - 0.4mm dia enamelled copper wire	5 metres	For transmitter and receiver coils
Resistor 1k	1	For current limiting
Transistor BC547	1	For switching
LED	1	For half-wave rectifier
USB cable	1	For charging mobile
9V battery with connector	1	For transmitter power supply



Fig. 1: Coils' winding



Fig. 2: The transmitter and receiver coils

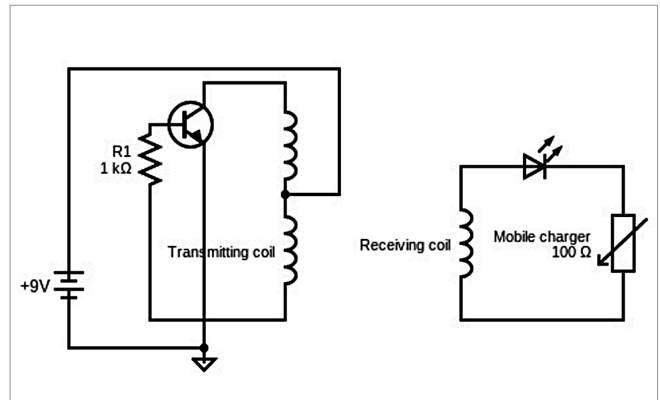


Fig. 3: The circuit diagram



Fig. 4: Author's prototype